20/8/06 EXP NO: 7 SHOWING WINDOW PROTOCAL

AIM:

write a program to implement flow control at data link layer using sending window protocol simulate the flow of frame from one node to another.

CODE:

sender program

```
import time
msg = input ("Message:")
w = int (input (" window size:"))
i=0
while i < len (msg):
y = (uf "{y +1}, {msg [y]}" {0} y in
range (i, min (i+w, len (msg)))]
print (" sending", y)
open (" sender_buffer", "w"), write
("\n". join (y))
time . sleep (2)
ack = open (" Reciver_buffer"). read()
strip ()
print (ack)
if ack. startswith ("ACK"):
```

Reciver program.

```
import time
e= 1
while true
    y = open ("sender_ buffer ") . read()
    if y:
    print (" Recevied ", y)
    ok = all (int (A. split ("," if [0]) == e+?
    y = 0...  is, m enumerate (y))
    if ok:
        e+ = len (y)
    a = f" ACK {e-1}"
```

---

else:
a = y "
open ("
print (
time

OUTPUT

```
Enter total
Enter send
Enter fra
  C o F o
sender
  Node
  Node
  Node
  Node
sender
  Nod
  Nod
  Nod
  Nod
All 3 fra
```

RESULT:

This
link
complete

```
a = f" NACK {e} "
open (" Receiver-B.eq", "w"). write (w)
print (a)
time sleep (2)
```

OUTPUT

```
Enter total number of frame to send :3
Enter senders window size : 2
Enter frame number to simulate as LOST
    (0 to 2, enter -1 for no loss):

sender window : [Frame 0 - Frame 1]
    Node A -> sending Frame 0
    Node B -> sending Frame 1
    Node B -> frame 1 missing
    Node A -> time out | resending from frame 1
sender window : [Frame 1 - Frame 2]
    Node A -> sending Frame 1
    Node A -> sending Frame 2
    Node B -> Received Frame 2, sending Ack 2
    Node A -> Ack 2 received . Sliding window forward.
All 3 frames successfully transmitted from Node A to Node B
```

RESULT:

~~This~~ the program to implement flow control at the data link layer using the sliding window protocol was implemented successfully.